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Getting started with eBPF

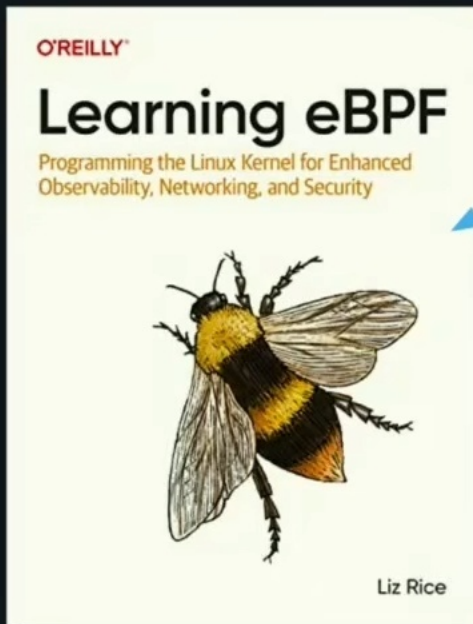


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Interactive labs

isogo.to/ebpf-vancouver

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What is  **eBPF** ?

extended
Berkeley
Packet
Filter

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What is  eBPF ?

Makes the kernel **programmable**

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Dynamically change kernel behaviour

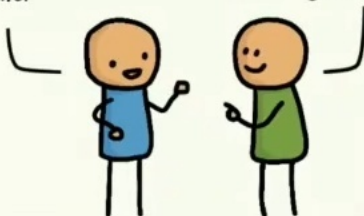
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Application Developer:

I want this new feature to observe my app



Hey kernel developer! Please add this new feature to the Linux kernel



OK! Just give me a year to convince the entire community that this is good for everyone.

1 year later...

I'm done. The upstream kernel now supports this.



But I need this in my Linux distro



5 years later...

Good news. Our Linux distribution now ships a kernel with your required feature



OK but my requirements have changed since...



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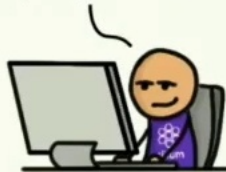
Application Developer:

i want this new feature
to observe my app



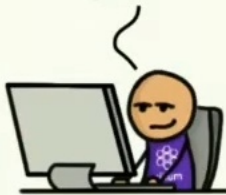
eBPF Developer:

OK! The kernel can't do this so let
me quickly solve this with eBPF.



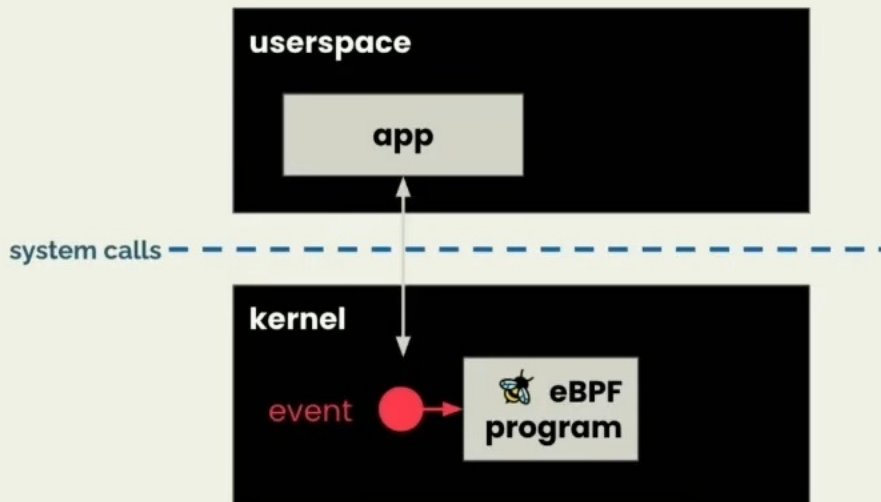
A couple of days later...

Here is a release of our eBPF project that has this feature
now. BTW, you don't have to reboot your machine.



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Run custom code in the kernel



Triggering eBPF programs

eBPF programs can be attached to different events

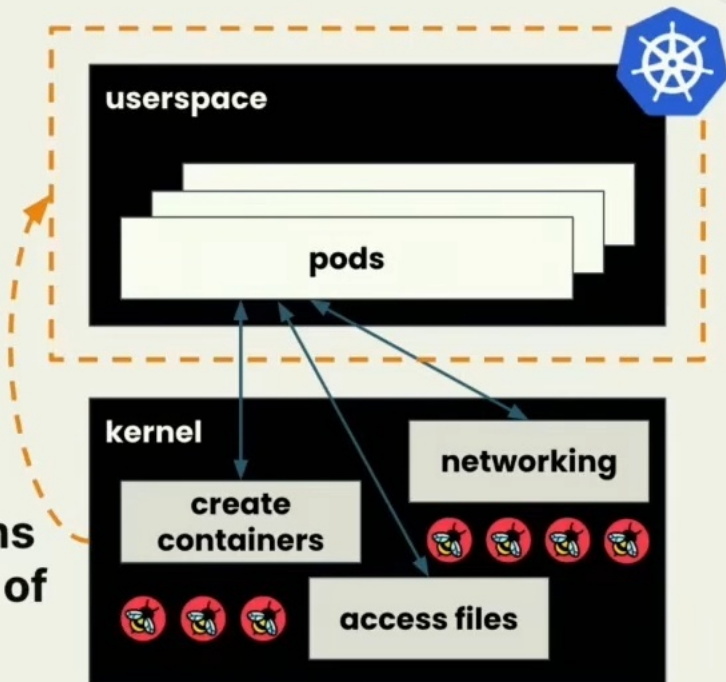
- Kprobes
- Uprobes
- Tracepoints
- Network packets
- Linux Security Module
- Perf events
- etc...

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Programmable kernel in Kubernetes

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eBPF programs
can be aware of
everything



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Lab: Hello World

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Examples from Chapter 2:

`cd learning-ebpf/chapter2`

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eBPF Hello World

Context depends
on event

```
SEC("kprobe/sys_execve")
```

+ userspace code to load eBPF
program

```
int hello(void *ctx)
```

```
{
```

```
    bpf_printk("I'm alive!");
```

```
    return 0;
```

```
}
```

Where do trace
messages go?

Info about process that
called execve syscall

```
$ ./hello.py
```

```
bash-20241
```

```
[004] d... 84210.752785: 0: I'm alive!
```

```
bash-20242
```

```
[004] d... 84216.321993: 0: I'm alive!
```

```
bash-20243
```

```
[004] d... 84225.858880: 0: I'm alive!
```

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bpf_trace_printk()

Example of an eBPF *helper function*

Always writes to `/sys/kernel/debug/tracing/trace_pipe`

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eBPF virtual machine

- Virtual machine implemented in software
- Executes eBPF bytecode instructions ([Unofficial eBPF spec](#))
- 10 general purpose registers

Instruction: `dst = src`

`0: (b7) r1 = 12345678`

dst is Register 1

src is a value

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ELF object file

eBPF opcodes
eBPF maps

userspace

kernel

BPF verifier

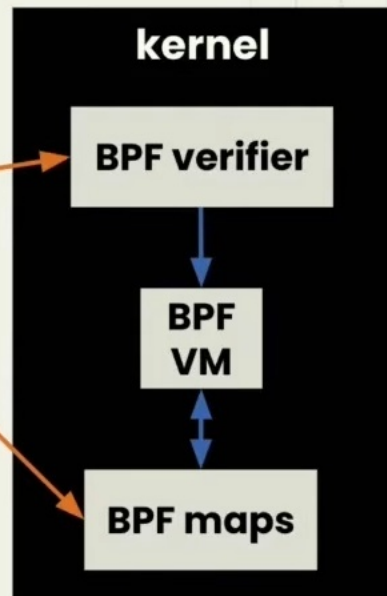
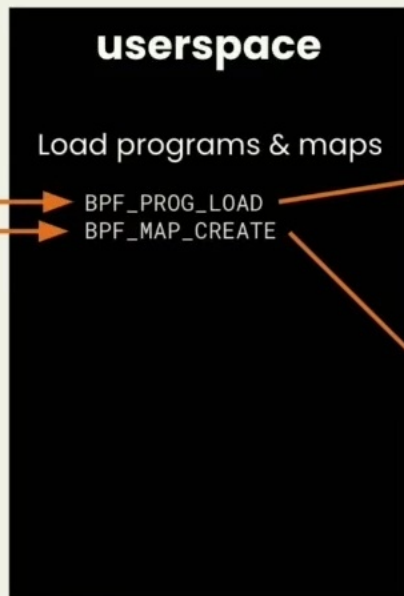
**BPF
VM**



BPF maps

syscalls

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syscalls

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ELF object file

eBPF opcodes
eBPF maps

userspace

Load programs & maps

BPF_PROG_LOAD
BPF_MAP_CREATE

Attach programs
to events

kernel

BPF verifier

BPF
VM

BPF maps

syscalls

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ELF object file

eBPF opcodes
eBPF maps

userspace

Load programs & maps

BPF_PROG_LOAD
BPF_MAP_CREATE

Attach programs
to events

Read / write maps

BPF_MAP_GET_NEXT_KEY
BPF_MAP_LOOKUP_ELEM
BPF_MAP_UPDATE_ELEM
BPF_MAP_DELETE_ELEM

kernel

BPF verifier



BPF
VM



BPF maps



syscalls

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Lab: bpftool and bytecode

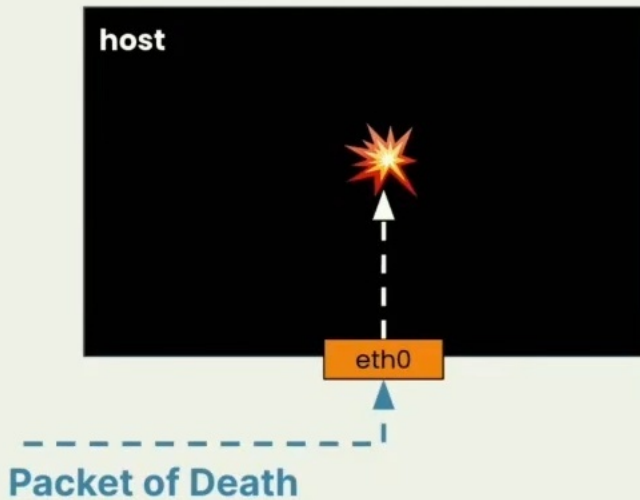
Lab: bpftool and maps

`cd learning-ebpf/chapter2`

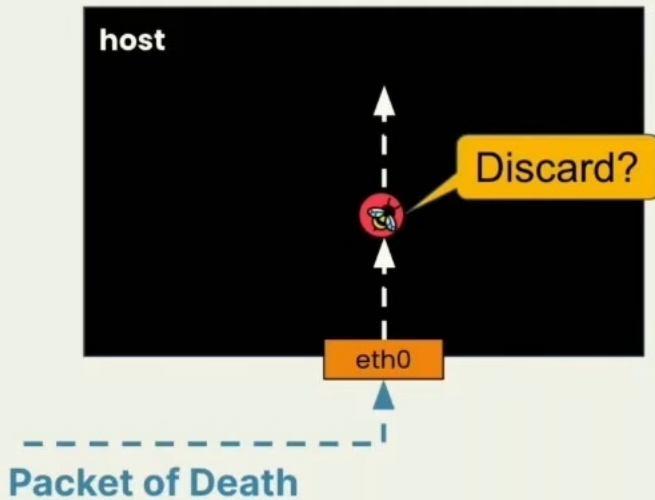
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eBPF for networking

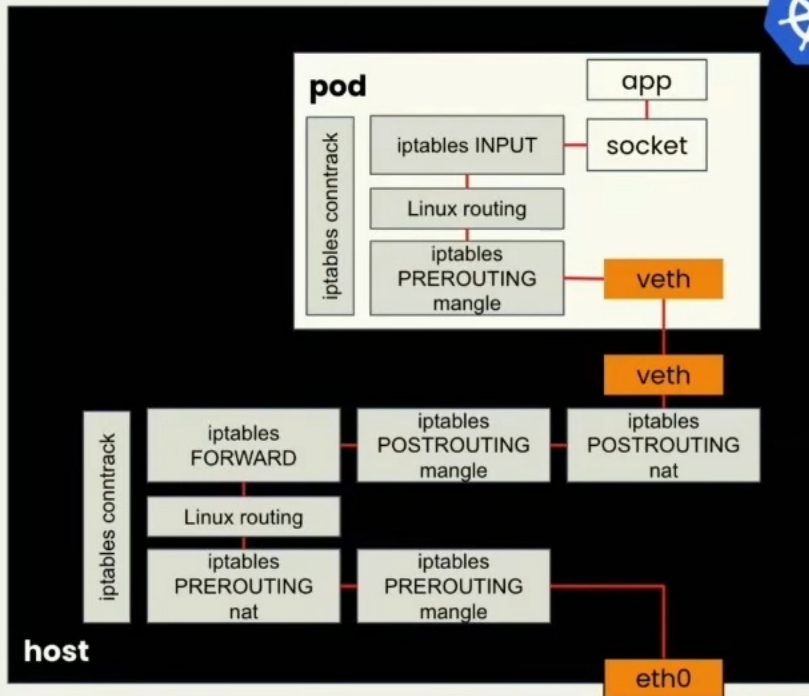
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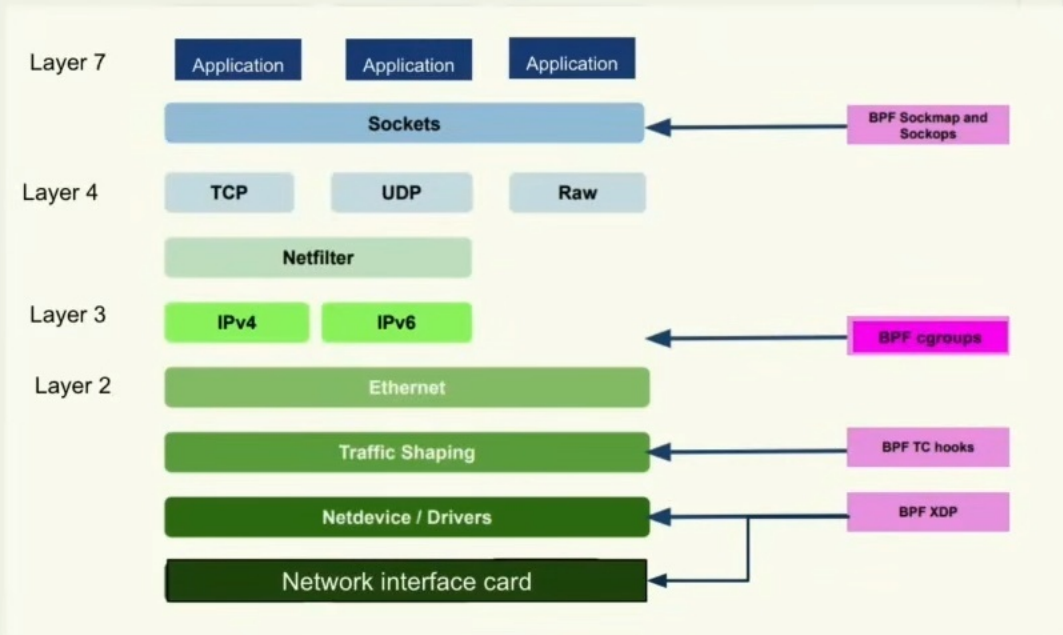
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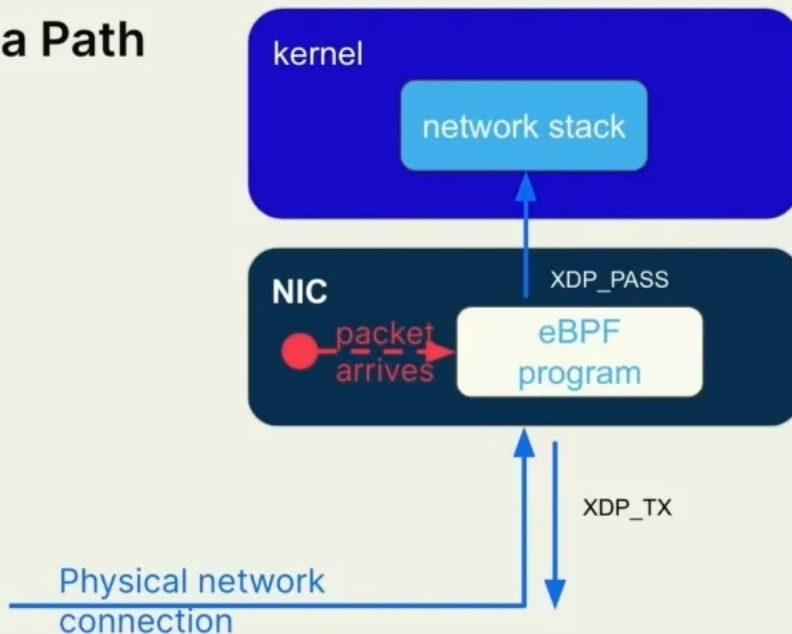


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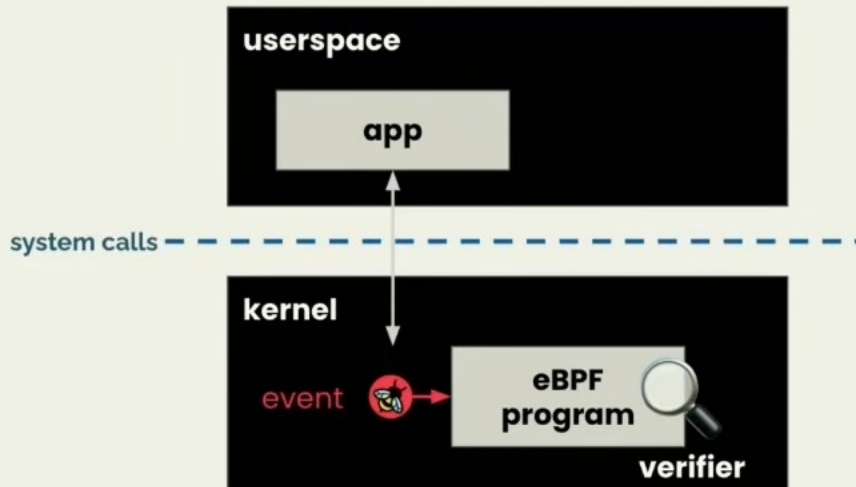
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eXpress Data Path



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eBPF code has to be safe



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Lab: eBPF Verifier

`cd learning-ebpf/chapter6`

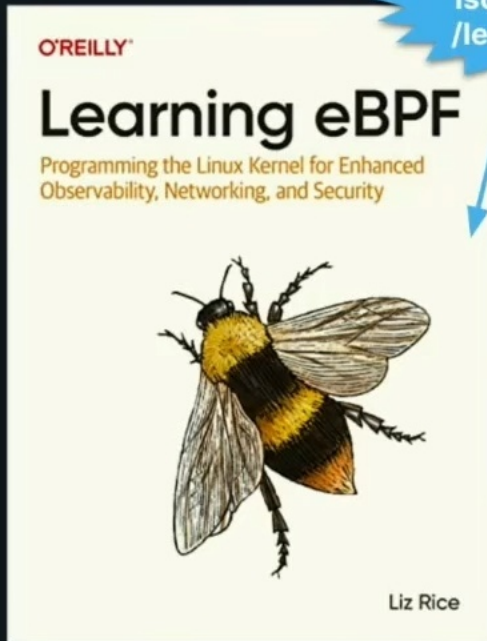
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Thank you

 [lizrice/learning-ebpf](https://github.com/lizrice/learning-ebpf)

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